Where do libraries come from, and where do they stop helping?

Bootstrapped libraries, documentation, evidence selection, and boundaries

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In one sentence

A 4-bit 8B model wrote a verified library for a new benchmark from three examples, and a 0.6B caller that never saw the benchmark uses it correctly to 1M tokens; the useful capacity is in the proposer and in choosing evidence, not in readers, recursion or larger controllers; a retrieval library also beats direct reading on LongBench-v2.

Libraries can be bootstrapped: an 8B wrote MRCR's library.

MRCR held out (8k to 1M tokens)	2 needles	4 needles
Direct answer, 4B	0.23	0.15
RLM, untrained 4B, no library	0.18	0.11
8B-written library, run alone	1.000	1.000
0.6B caller + library	**1.000**	**1.000**
Untrained 4B + library	0.95	0.88

64 rows per column. The 0.6B caller never saw MRCR. The proposer saw the contract and three solved examples; candidates run on 30 teaching tasks; the best scored 30/30 and was reviewed before use (40 lines of parsing, nothing copied from the examples).

Who can write it (seeds accepted)

4B Instruct: 0 of 1
4B, thinking off / on: 0 of 1 / 0 of 2
8B, thinking off: 0 of 2
8B, thinking on: **2 of 5**
4B after SFT + RL: 0 of 4; test-time RL on MRCR: 0 in 1,920 samples

Caveat: MRCR is solvable by a short parser, so this measures program discovery, not reading. We found no published note saying so.

Ledger E21, E21b; runs e21-*. MRCR: openai/mrcr (MIT), official SequenceMatcher grader.

Discovery on 37 generated families: thinking, scale, and training.

Proposer	Solved / 37	Tokens
4B Instruct (2 seeds)	5, 4	4k
8B, no thinking	4	4k
4B, thinking (30 done)	13 / 30	
8B, thinking	**20**	49k
4B + SFT (2 seeds)	9, 7	3k
4B + SFT + RL (2 seeds)	**16, 14**	2k

Tokens: proposer output per family. Families vary record format (pipe, kv, csv, prose, XML-like, chat), question and answer format; held-out families use a new format or a new question.

Training the proposer: SFT on the 8B's verified code, then 40 steps of RL with the verified teaching score as reward: 9 to 30 of 74 attempts ($p < 0.001$); string formats 1 to 22 of 42. MRCR still fails.

Helper ceiling: the 8 families that need the helper's judgment score 0.47 to 0.83, not 0.9: the 4B labels items at 0.973, and an exact total over 30 items fails often.

Ledger E26, E26b, E26c; runs e26-disc-*, e26c-proposer-rl. Generator: family_gen.py (procedural, answers exact).

One line of documentation decides whether an untrained caller uses a library.

MRCR, untrained 4B caller, docstring	Score
Signature only	0.18
Description + signature (as the 8B wrote it)	0.59
Signature + usage line	0.95
Description + signature + usage line	0.95
Hand-written library, usage line removed	0.22
Hand-written library, as written	0.95

Usage line: `Typical use: FINAL_TEXT(answer_mrcr(request)).`

Without it the untrained 4B hands the whole input to a sub-model (overflowing at 128k and above) or retypes the routine's output by hand.

The distilled 0.6B scores 1.000 without the line; the untrained 0.6B fails either way (0.21, 0.13). Robustness comes from distillation, not size. The bootstrapper now inserts the line.

Ledger E25c, E25d; runs e25c-*, e25d-*. One family, one library, one seed per arm.

MuSiQue in long haystacks: capacity belongs in choosing evidence.

Answer F1 (90 to 180 held-out questions per row)	8k	32k	128k	tokens
Direct answer, 4B	0.52	0.25	overflow	
RLM without a library, depth 0 / depth 1	0.49 / 0.51	0.25 / 0.25	overflow	
Library that hands each hop to a child RLM	0.29	0.29	0.34	3 to 4 times
0.6B caller + filter, BM25, read	0.52	0.31	0.46	57k
0.6B caller + two-round retrieval	**0.53**	**0.39**	**0.50**	10k

What helps: a second, targeted retrieval (BM25 for the question, the helper names the missing fact, BM25 again, one read): +0.058 over BM25 alone, +0.045 over filter+BM25 at a sixth of the tokens.

What does not: recursion; an 8B reader (+0.01; gold paragraphs reach 0.63 to 0.67 with either reader); more sampling (majority of 3 at 88k tokens: 0.22 vs 0.35 for the library at 32k).

Ledger E22c to E22f, E23, E25e, E28. MuSiQue (CC BY 4.0) dev questions in haystacks of other questions' paragraphs.

External validity, and what we withdrew.

LongBench-v2 (150 questions, chance 0.25)

Direct, truncated to 48k tokens: 0.367 (42k tokens)
Map-reduce notes library: 0.360 (366k tokens)
Retrieve-then-read library: 0.480 (10k tokens)
RLM without a library: 0.120

Selecting evidence beats summarizing everything, as on MuSiQue.

Withdrawn: "the reader is distractor-limited"; length routing (pre-registered: the ordering reversed on fresh questions); "libraries stop helping on LongBench-v2" (it was the map-reduce routine).

Method lessons

- Verify under the semantics you run under: the kernel's `ask` raises on a cut-off reply, scripted harnesses returned partial text (one routine crashed in 46 of 90 caller runs).
- Verification must span deployment lengths: 8k-only teaching selected a pass-through routine that scores 0 at 128k.
- Routine evolution (GEPA-style): prompt rewrites by the 8B hurt; numbers helped (+0.048, ns); random search found nothing.

Ledger E32, E32b, E31, E22b, E24, E24c. LongBench-v2: zai-org (Apache-2.0).

What this changes, and what to test next.

The thesis, sharpened. For small recursive models, capability lives outside the weights: verified routines (bootstrapped by a capable proposer, or by a small one trained on verified code and rewarded by verification), documentation a thin caller can act on, and evidence selection inside the routine. A 0.6B caller with a retrieval library beats a 4B reading the document on MuSiQue and on LongBench-v2.

Next

1. Train the proposer on families that need the helper or a search strategy: trained only on code-solvable families, the 4B became a parser writer and ignores retrieval primitives.
2. Primitives plus proposer training for MuSiQue-like families (the 8B composes BM25 into a working routine on 1 of 2 seeds).
3. Selection memory with pooled priors, since 30-question cells choose by noise.
4. A short note on program-solvable long-context benchmarks (MRCR; candidates: Michelangelo, RULER).

Appendix: protocol and provenance.

Models: Qwen3-4B-Instruct-2507 (helper, caller), Qwen3-0.6B + LoRA (distilled caller), Qwen3-4B hybrid and Qwen3-8B-AWQ (proposers), all served with vLLM 0.27.1 on two A100 MIG 7g.40gb slices.

Data: OpenAI MRCR (MIT; within-bin split, 30 teaching rows, 64 held out per needle count); MuSiQue v1.0 (CC BY 4.0; teaching from train, held out from dev; two samples of 90); LongBench-v2 (Apache-2.0, revision 2b48e494); generated families (family_gen.py, exact answers).

Statistics: paired differences with standard errors over all questions; exact McNemar for family counts; the routing test was pre-registered.

Every number above has a ledger row in docs/RESULTS.md and a run directory in the run store; the lab notebook (docs/notebook/2026-10-19-...) records failures and corrections in time order.